

Practical Demo: Vasicek Rate Scenarios and Long-Run Behavior

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

How much of next year's funding-rate uncertainty comes from mean reversion, and how much comes from volatility?

Purpose and Assumptions

The Vasicek model is

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t.$$

This demo estimates its parameters from a dated monthly U.S. effective federal funds-rate history, simulates scenarios, checks long-run moments, and uses the fitted distribution for a one-year risk decision. The embedded observations are monthly averages for January 2023 through June 2025, recorded in percent and converted to decimals. Source: [Federal Reserve Bank of St. Louis, Effective Federal Funds Rate \(FEDFUNDS\)](#). Parameters are constant; estimation error, regime shifts, jumps, and term premia are ignored. This short sample is pedagogical and should not be treated as a robust trading calibration.

1 Step 1: Record the observed rate history

The treasury team's question: how uncertain is next year's floating interest expense? We will first fit historical persistence, then simulate monthly resets, then turn those resets into dollars. A good fit to history is not evidence that the next policy regime will behave the same way.

	Effective federal funds rate
2025-02-28	0.0433
2025-03-31	0.0433
2025-04-30	0.0433
2025-05-31	0.0433
2025-06-30	0.0433

The series shows tightening, a long plateau, and initial easing. It is too short and policy-driven for a robust structural calibration, but it lets us show the entire estimation workflow without hiding the model's limitations.

2 Step 2: Estimate Vasicek parameters

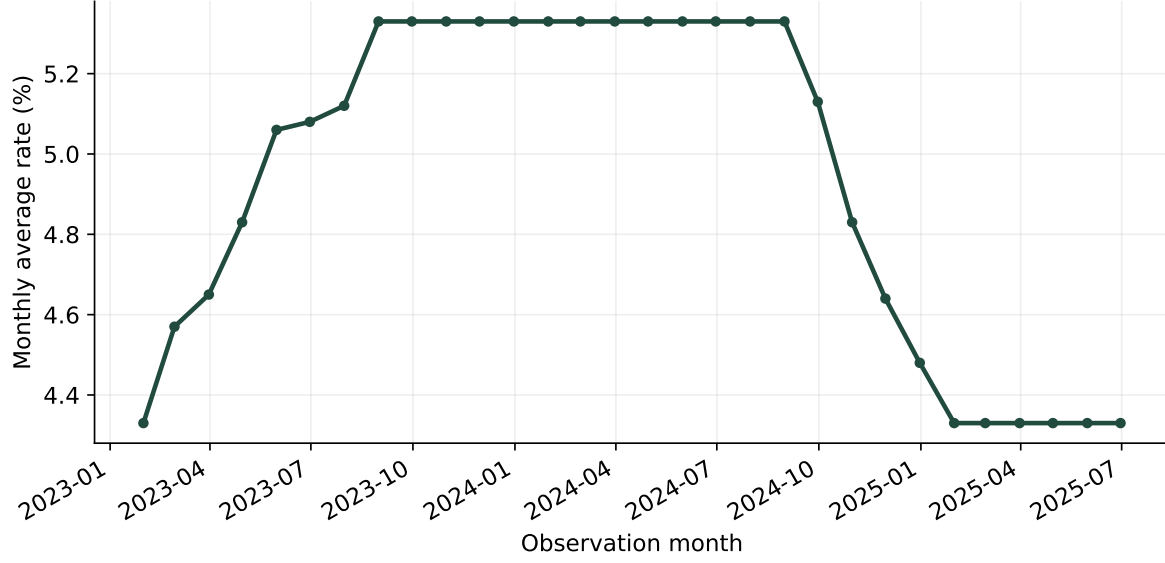


Figure 1: Monthly effective federal funds rates, January 2023–June 2025; the calibration sample is short and policy-driven.

The exact monthly transition can be written as an AR(1):

$$r_{t+\Delta} = \alpha + \phi r_t + \varepsilon_{t+\Delta}, \quad \phi = e^{-\kappa\Delta}, \quad \alpha = \theta(1 - \phi).$$

After OLS estimation,

$$\hat{\kappa} = -\frac{\log \hat{\phi}}{\Delta}, \quad \hat{\theta} = \frac{\hat{\alpha}}{1 - \hat{\phi}}, \quad \hat{\sigma} = s_\varepsilon \sqrt{\frac{2\hat{\kappa}}{1 - \hat{\phi}^2}}.$$

Here $\Delta = 1/12$ year, κ is in 1/year, and σ is in decimal rate units per $\sqrt{\text{year}}$.

Current rate	0.0433
Estimated monthly persistence phi	0.9578
Mean-reversion speed kappa (1/year)	0.5173
Long-run mean theta	0.0495
Volatility sigma (rate/sqrt(year))	0.0043
Mean-reversion half-life (years)	1.3399
Long-run standard deviation	0.0042
dtype: float64	

- κ controls how quickly shocks decay; the half-life is $\ln(2)/\kappa$.
- θ is the long-run mean under the chosen probability measure.
- σ controls the size of random innovations and the long-run dispersion.

3 Simulate Monthly Paths

i What the regression does—and does not—fit

Each row asks whether a high rate this month tends to remain high next month. The slope ϕ measures persistence; ϕ near one means slow mean reversion. The residual is the part not explained by that straight line. Inspect it before accepting the normal, independent-shock assumption. Monthly average FEDFUNDS is a proxy for a short rate, not an observed continuously evolving spot rate.

The exact Vasicek transition is normal:

$$r_{t+\Delta} = \theta + (r_t - \theta)e^{-\kappa\Delta} + \sigma\sqrt{\frac{1 - e^{-2\kappa\Delta}}{2\kappa}}Z.$$

4 Distribution and Negative Rates

Five-year mean	0.0492
Five-year standard deviation	0.0042
5th percentile	0.0421
Median	0.0492
95th percentile	0.0560
Probability of negative rate	0.0000
dtype: float64	

Vasicek rates are normally distributed, so negative rates are possible. That can be useful in markets where negative policy rates are plausible, but it may be undesirable for other applications. A positivity-preserving model such as CIR changes this feature, at the cost of a different distribution and dynamics.

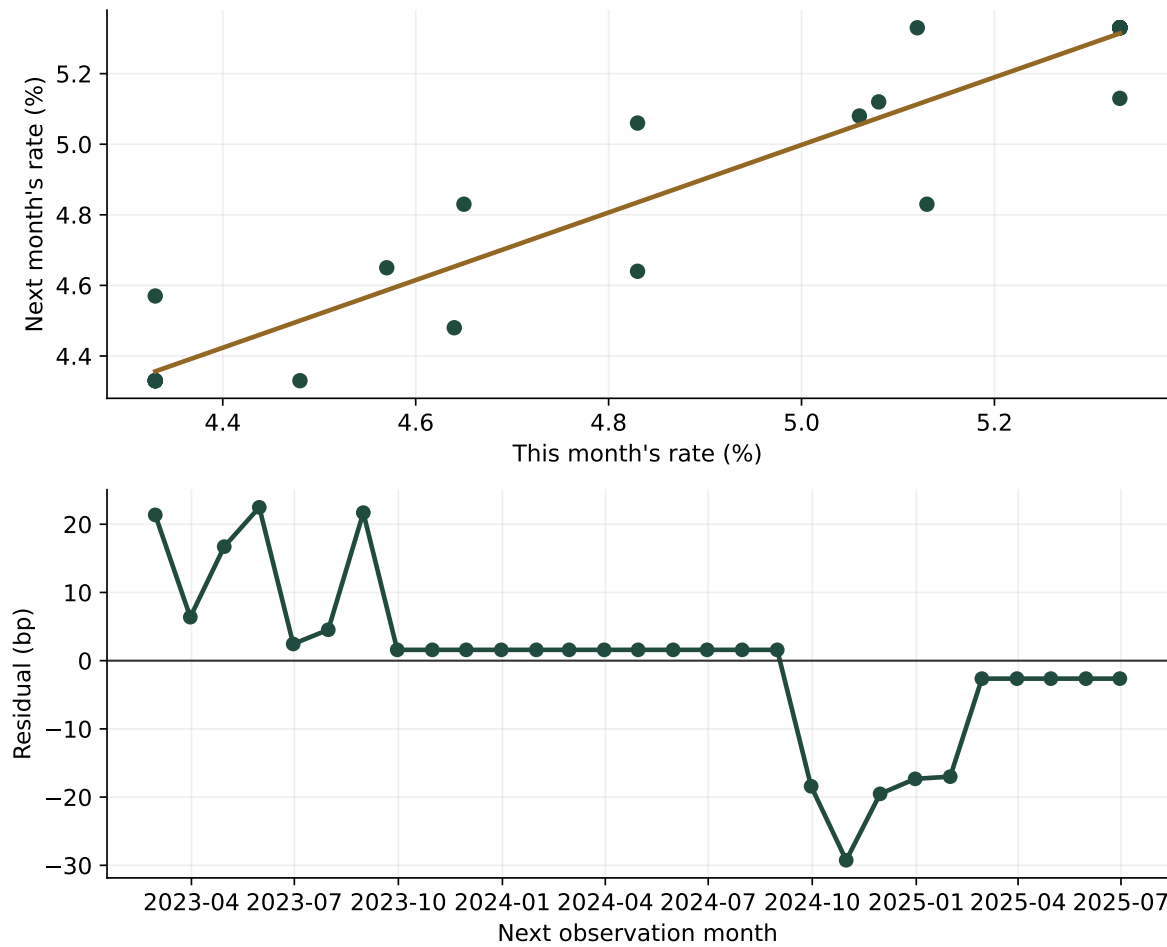


Figure 2: AR(1) fit and residuals, using 29 monthly transitions. Residual patterns challenge the constant-parameter assumption.

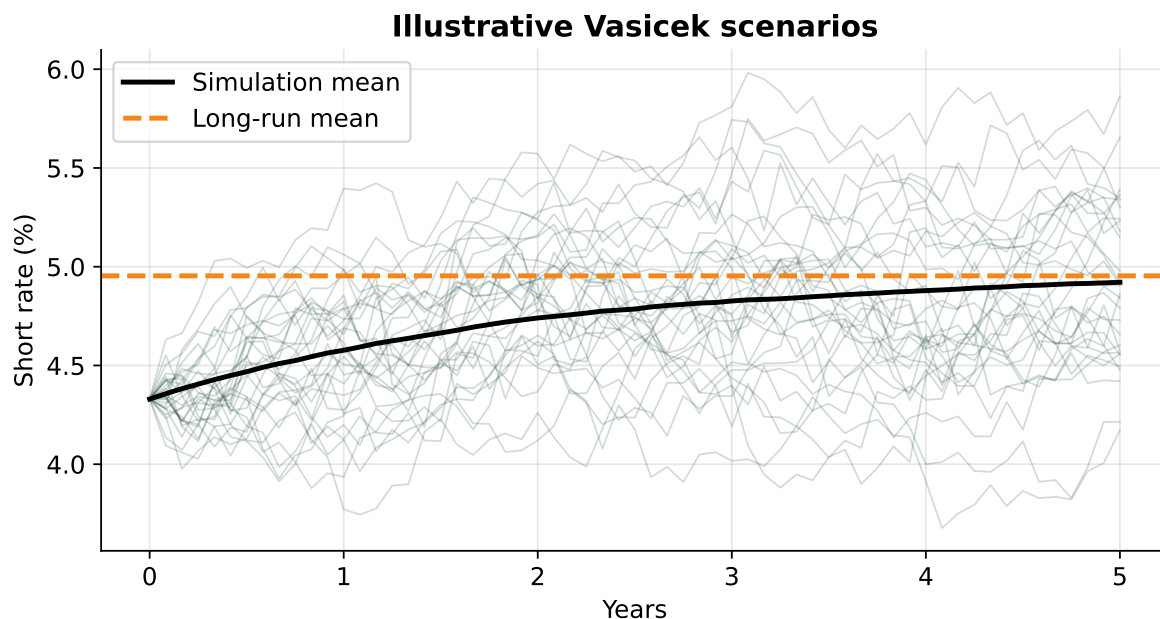


Figure 3: Vasicek paths fluctuate randomly but revert toward the long-run mean

5 Use the fitted distribution in a funding decision

Suppose a treasury team wants a one-year planning range for a floating-rate funding cost tied to the overnight rate. Simulate 50,000 one-year endpoints and report the central 90% range and the probability that the rate exceeds 5%. This applies the fitted **physical-measure** model; it is not a risk-neutral bond price.

5th-percentile funding rate (%)	4.023
Median funding rate (%)	4.581
95th-percentile funding rate (%)	5.137
Probability funding rate exceeds 5%	0.107

dtype: float64

The 95th percentile can serve as a planning stress, while the exceedance probability helps size an interest-expense buffer. Parameter uncertainty and policy regime changes should be added as separate stresses.

6 Model Scenario Versus a Historical Shock

6.1 Translate the path into an interest-expense budget

The year-end rate is a **reset-rate** scenario, not the year's total interest expense. A \$10 million borrowing with monthly resets pays interest throughout the year. For a constant 100 bp loan margin, the simplified annual cash cost is

$$\text{Interest} = 10,000,000 \sum_{k=0}^{11} (r_{k/12} + 0.01)/12.$$

We use the rate at the start of each monthly period, an equal-month accrual convention, and no compounding of unpaid interest. The loan is illustrative; the historical FEDFUNDS sample supplies the fitted dynamics.

Flat current-rate budget (\$)	533000.00
Simulated median expense (\$)	545538.37
95th-percentile expense (\$)	577428.31
Probability expense exceeds flat budget	0.74
dtype: float64	

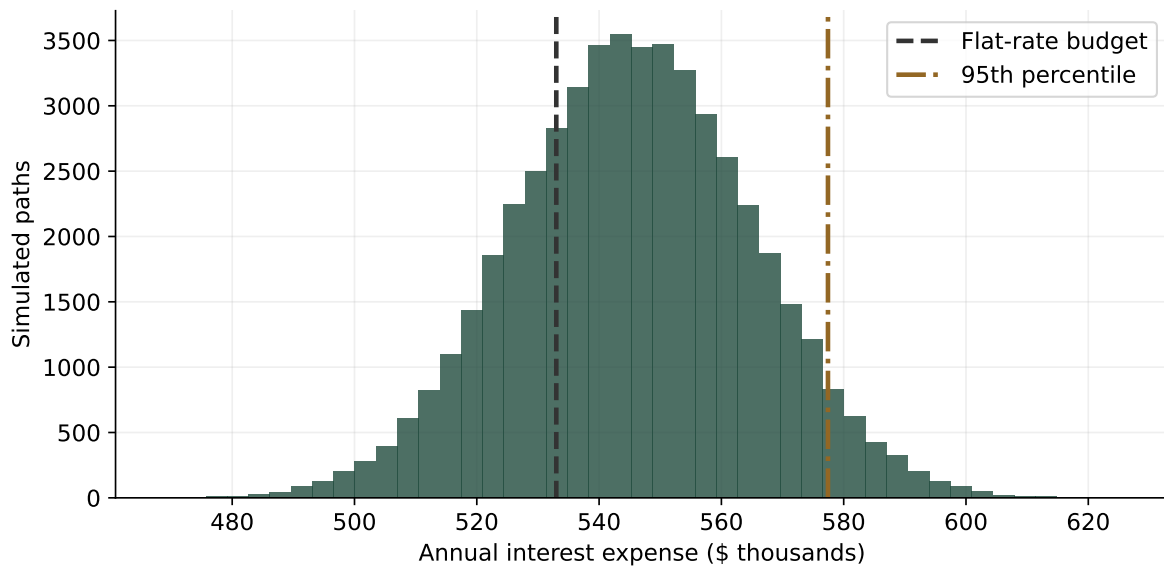


Figure 4: Simulated annual interest expense, not a confidence interval or a guaranteed maximum.

💡 Try a treasury decision

Double the notional, then raise the loan margin by 50 bp. The first change scales the entire expense distribution; the second adds \$50,000 for the original \$10 million loan. Neither changes the fitted rate process. Recommend a cash buffer, then explain why a policy-regime stress should accompany it.

Use the largest one-month decline in the compact history as a simple historical shock, and compare it with the model's one-month distribution.

```
Largest historical monthly decline (bp)    -30.00
Model mean one-month change (bp)          2.67
Model 5th-percentile change (bp)          -17.46
Model 1st-percentile change (bp)          -25.37
dtype: float64
```

The historical scenario preserves an observed move but supplies only one outcome. The Vasicek simulation supplies a distribution and mean reversion, but depends on model specification and parameters. Risk management should compare both rather than treating either as truth.

7 Validate Long-Run Moments and Units

The stationary Vasicek distribution has

$$E[r] = \theta, \quad \text{Var}(r) = \frac{\sigma^2}{2\kappa}.$$

```
Theoretical long-run mean          0.0495
Simulated 20-year mean             0.0496
Theoretical long-run standard deviation  0.0042
Simulated 20-year standard deviation  0.0043
Decimal 0.0001 equals one basis point    True
dtype: object
```

8 Financial Interpretation

Mean reversion makes the expected rate move toward θ , while volatility keeps future rates dispersed. κ determines how quickly today's rate loses influence; σ determines scenario width. The model is a disciplined scenario generator, not a forecast guarantee. Calibration window, probability measure, structural breaks, and the normal model's negative-rate tail all affect how its output should be used.

9 Interactive explorer

Change the forecast horizon, keeping the historical fit fixed. The mean moves toward the estimated long-run rate while dispersion approaches a limit. The 5th–95th percentile band is conditional model uncertainty, not parameter uncertainty and not a confidence interval for the mean.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

10 Mini Project

Project brief. Use the historical fit in this demo to compare the conditional mean and 5th–95th percentile short-rate range at 0.5, 1, and 2 years. Translate each terminal rate into an annualized interest run rate on \$10 million floating-rate borrowing with a fixed 100 bp spread. Explain why this is not the distribution of total interest paid over the horizon. Report one reason the fitted historical model is unsuitable for pricing an option without further work.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

	Horizon (years)	Mean rate (%)	5th rate (%)
0	0.5	4.4722	4.029
1	1	4.5819	4.022
2	2	4.7321	4.0802

	95th rate (%)	Mean annual run rate (\$)	95th annual run rate (\$)
0	4.9154	547,217.3884	591,538.696

	95th rate (%)	Mean annual run rate (\$)	95th annual run rate (\$)
1	5.1419	558,194.3734	614,188.695
2	5.384	573,212.9938	638,401.237

The range widens toward the stationary distribution while the conditional mean approaches theta. Multiplying the terminal rate plus spread by notional gives an annualized expense at that future instant. Total interest requires integrating or summing rates along each path. Historical parameters describe a physical-measure approximation; option pricing needs risk-neutral dynamics and a fit to current pricing inputs. This short, policy-driven sample also understates parameter and regime uncertainty.